Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Hammer, Daniel Ph.D.		
Project Title:	Polymosomes: Tough Giant Vesicles from Block Copolymers		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Polymers		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	03/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Discher, D. (University of Pennsylvania) Bates, F. (University of Minnesota)		
Grant/Contract No.:	NAG8-1683		
Performance Goal No.:			
Performance Goal Text:			
	Polymersomes are vesicles that can be made of a synthetic material – a block copolymer. The strategy is to use polymers that are amphiphilic, and will form lamellar phases in water, to mimic the formation of a biological membrane. The properties of the membrane can be set through synthetic control of block copolymer chemistry. The architecture and chemistry of the blocks, making di- and tri-block copolymers, as well as making polymers that can be crosslinked or will degrade upon cue, will allow us to make a wide variety of vesicles with different physical properties. The properties are measured using micropipette aspiration, a technique ideal for measuring the materials properties of micron-scale objects. Also, materials (fluorescent dyes, oxygen binding molecules) can be incorporated within the interior of the vesicles. The combination of encapsulating oxygen binding molecules and crosslinking the vesicle membrane can lead to the formation of an artificial erythrocyte, which can store oxygen. Also, the polymersomes display interesting shape transitions that are likely affected by gravity and the material properties of the membranes, which can be explored		

Task Description:	further using encapsulants and solvents of different densities and membranes made of different block co-polymer chemistries and architectures. The task objectives are to make vesicles from block copolymers, including di- and tri-block copolymers; crosslink the vesicles to make solid-like structures that mimic the erythrocyte material properties; encapsulate materials into the interior of the vesicles; and to determine if the shape response of the vesicles is affected by gravitational forces. The significance of this work is that it will lead to the development of artificial cells that can be used and studied in space applications. For example, it may be possible to make an artificial red blood cell that can be freeze dried and stored for long-term space exploration. Also, studying the fundamental behavior of polymer vesicles – as a function of polymer chemistry – will tell us how cells will behave in microgravity. Finally, the effect of microgravity on vesicle assembly and behavior may suggest new ways to make and assemble materials with novel optical, biological and rheological properties in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period. [Ed. note (2/6/2020): Some publications added that result from work on this project.]
Bibliography Type:	Description: (Last Updated: 02/06/2020)
Articles in Peer-reviewed Journals	Bermudez H, Aranda-Espinoza H, Hammer DA, Discher DE. "Pore stability and dynamics in polymer membranes." EPL (Europhysics Letters). 2003 Nov;64(4):550-6. https://doi.org/10.1209/epl/i2003-00264-2 , Nov-2003
Articles in Peer-reviewed Journals	Milam VT, Hiddessen AH, Crocker JC, Graves DJ, Hammer DA. "DNA-driven assembly of bidisperse, micron-sized colloids. " Langmuir. 2003 Nov;19(24):10317-23. https://doi.org/10.1021/la034376c , Nov-2003